

AlphaForge - AI-Powered Financial Intelligence Platform

Subject – PGM

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CERTIFICATE

This is to certify that the Mini Project entitled “**AlphaForge**” is a bonafide work of **Mansi Jadhav (24101C2003)**, **Atharva Gandhe (23101C0009)**, **Tanishq Chavan (23101C0008)**.

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Contents

Section No.	Title	Page No.
-	Abstract	ii
-	Acknowledgments	iii
-	List of Abbreviations	iv
-	List of Figures	v
-	List of Tables	vi
-	List of Symbols	vii
1	Introduction	1
1.1	Problem Statement & Objectives	2
2	Literature Survey	11
2.1	Survey of Existing/Similar System	13
3	Proposed System	18
3.1	Introduction	19
3.2	Architecture / Framework	20
3.3	Algorithm and Process Design	21
3.4	Details of Hardware & Software	23
3.5	Experiment and Results	26
3.6	Conclusion and Future Work	30
4	References	32

Abstract

AlphaForge is a next-generation financial intelligence platform designed to provide explainable and data-driven insights into financial markets using Probabilistic Graphical Models (PGMs). Unlike traditional black-box machine learning approaches, this system leverages Bayesian Networks to model causal relationships between market indicators and generate transparent predictions. The platform integrates real-time feature engineering, advanced analytics, and interactive visualization tools to enhance decision-making in trading and financial analysis. It also incorporates backtesting mechanisms and performance evaluation metrics to validate predictive models. With a modern frontend built using Next.js and a robust backend powered by FastAPI, AlphaForge delivers a scalable and efficient solution for financial forecasting and analysis.

Furthermore, the system incorporates advanced analytical capabilities such as feature contribution analysis, scenario simulation, and failure case evaluation, enabling users to understand the impact of individual market indicators on prediction outcomes. The integration of a Bayesian Network allows the system to perform probabilistic inference, making it possible to compute conditional probabilities of future returns based on current market conditions.

In addition, AlphaForge provides a comprehensive backtesting framework that evaluates multiple trading strategies using performance metrics such as accuracy, Sharpe ratio, win rate, and maximum drawdown. This helps in validating the effectiveness of predictive models in real-world scenarios. The platform also supports real-time data processing through efficient feature engineering pipelines, ensuring up-to-date insights for users.

Acknowledgment

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List of Abbreviations

- PGM – Probabilistic Graphical Model
- API – Application Programming Interface
- UI – User Interface
- DAG – Directed Acyclic Graph
- RSI – Relative Strength Index
- MACD – Moving Average Convergence Divergence
- ATR – Average True Range
- AI – Artificial Intelligence

List of Figures

Figure 1: System Architecture of AlphaForge Platform

Figure 2: Market Dashboard displaying real-time market regime, volatility index, and active signals

Figure 3: Stock analysis with price trends and technical indicators (SMA, RSI, MACD)

Figure 4: Backtesting performance comparison of trading strategies

Figure 5: Directed Acyclic Graph (DAG) representing Bayesian Network

Figure 6: Bayesian Network structure validation and correlation analysis

Figure 7: Feature contribution analysis showing impact of indicators on prediction

Figure 8: Performance comparison of PGM model with baseline models

Introduction

AlphaForge is a financial intelligence platform that utilizes advanced AI techniques to analyze stock market data and generate predictive insights. The system is built to overcome the limitations of traditional black-box models by offering explainable predictions through Bayesian Networks. It integrates multiple components such as data ingestion, feature engineering, probabilistic modeling, and visualization into a unified platform. The application provides users with real-time analytics, scenario simulations, and model evaluation tools, making it suitable for traders, analysts, and researchers.

1.1 Problem Statement & Objectives

The financial market is highly volatile and complex, making accurate prediction and analysis challenging. Traditional machine learning models often lack transparency and fail to explain the reasoning behind predictions.

The main objectives of this project are to develop an explainable financial prediction system using Bayesian Networks, to provide real-time feature engineering and analytics, to enable scenario-based simulations for decision-making, and to evaluate model performance using statistical metrics. Literature Survey

The field of financial market prediction has been widely explored using various computational and statistical techniques. Traditional approaches rely on statistical models such as linear regression and time-series analysis, including ARIMA

models, which are effective for capturing trends and seasonality but fail to model complex nonlinear relationships between financial variables.

With the advancement of machine learning, models such as Support Vector Machines (SVM), Decision Trees, and Random Forests have been applied to stock market prediction. These models improve predictive performance by capturing nonlinear patterns in data. However, they still lack interpretability and do not provide clear reasoning behind predictions, which is a critical requirement in financial decision-making.

Deep learning techniques such as Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks have gained popularity due to their ability to model temporal dependencies in time-series data. These models achieve high accuracy but require large datasets and computational resources. Moreover, they act as black-box systems, making it difficult for users to understand how predictions are generated.

Probabilistic Graphical Models (PGMs), particularly Bayesian Networks, offer a promising alternative by representing variables and their dependencies using Directed Acyclic Graphs (DAGs). These models enable probabilistic reasoning and provide interpretable results. They can capture causal relationships and allow for inference under uncertainty, making them highly suitable for financial analysis.

Literature Survey

2.1 Survey of Existing/Similar System

Existing financial platforms such as algorithmic trading systems and stock analysis dashboards primarily focus on visualization and prediction accuracy. Many systems use technical indicators like RSI, MACD, and moving averages to generate trading signals. While these approaches are useful, they rely on predefined rules and do not adapt well to changing market conditions.

Machine learning-based systems improve prediction accuracy but often lack transparency. Deep learning models, although powerful, are computationally expensive and difficult to interpret. Some platforms provide backtesting features, but they do not integrate explainable AI techniques or probabilistic reasoning.

Recent research has explored the use of Bayesian Networks in financial prediction, highlighting their ability to model dependencies between variables and perform probabilistic inference. However, most implementations are limited to theoretical studies and lack practical, user-friendly interfaces.

AlphaForge addresses these limitations by integrating Bayesian Networks with real-time feature engineering, interactive visualization, and backtesting capabilities. It combines the strengths of probabilistic modeling and modern web technologies to provide a comprehensive and explainable financial intelligence platform.

Proposed System

3.1 Introduction

The proposed system, AlphaForge, is an advanced financial intelligence platform designed to provide accurate and explainable predictions for stock market analysis. The system leverages Probabilistic Graphical Models, specifically Bayesian Networks, to model the dependencies between various financial indicators such as momentum, volatility, and trend-based features. Unlike conventional machine learning models, AlphaForge focuses on capturing causal relationships between variables, enabling more meaningful and interpretable predictions.

The platform integrates multiple components including data ingestion, feature engineering, probabilistic inference, and visualization into a unified pipeline. Real-time financial data is collected from external sources and processed to generate technical indicators such as RSI, MACD, Bollinger Bands, and volatility measures.

These features are then discretized and used as inputs to the Bayesian Network, which performs probabilistic reasoning to estimate future market movements.

3.2 Architecture/ Framework

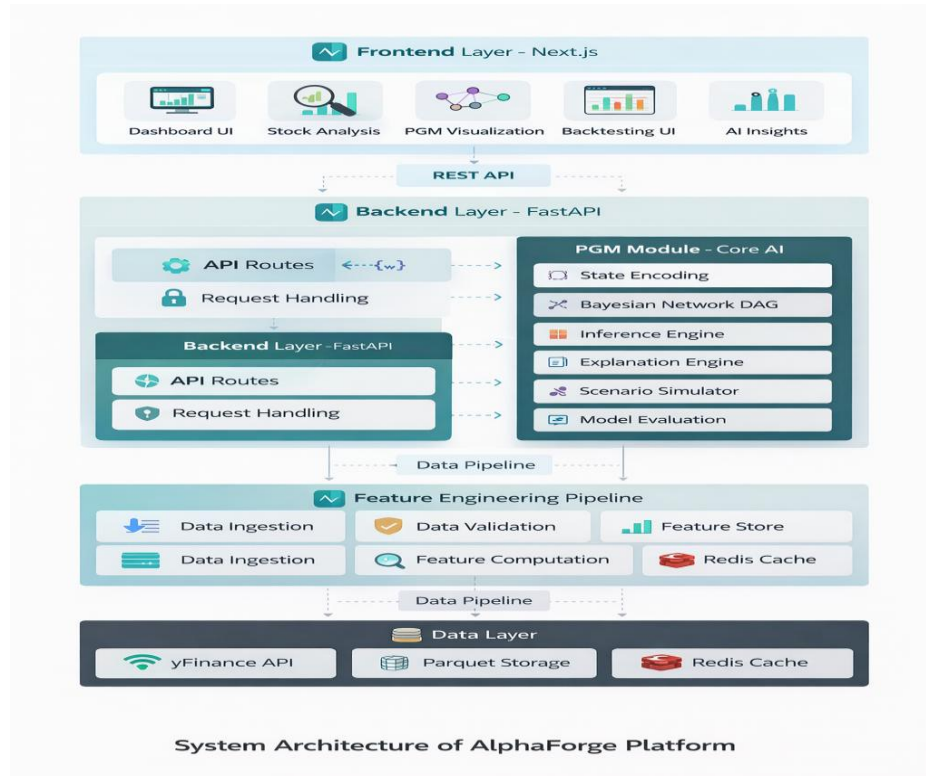


Figure no 1: **System Architecture of AlphaForge Platform**

The system follows a three-layer architecture consisting of frontend, backend, and data layers. The frontend is built using Next.js and provides dashboards, stock analysis, and visualization tools. The backend is implemented using FastAPI and handles API requests, feature processing, and model inference.

The core component is the PGM module, which includes state encoding, Bayesian Network structure, inference engine, explanation engine, and scenario simulation. The data layer includes yfinance API for stock data, Parquet files for storage, and Redis for caching.

3.3 Algorithm and Process Design

The AlphaForge system follows a well-defined pipeline that transforms raw financial data into explainable predictions using Probabilistic Graphical Models.

The overall process consists of multiple stages including data ingestion, preprocessing, feature engineering, model training, probabilistic inference, and evaluation.

Initially, historical stock market data is collected from external sources such as financial APIs. This data includes attributes like open price, close price, high, low, and trading volume. The collected data undergoes preprocessing steps such as cleaning, handling missing values, and normalization to ensure data quality and consistency.

In the next stage, feature engineering is performed to extract meaningful indicators from raw data. Technical indicators such as Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Bollinger Bands, Average True Range (ATR), and momentum scores are computed. These features help in capturing market trends, volatility, and price behavior.

Since Bayesian Networks operate on discrete variables, continuous features are converted into discrete states using discretization techniques such as quantile-based binning. This step transforms numerical values into categories like low, medium, and high, enabling efficient probabilistic modeling.

The discretized features are then used to construct a Bayesian Network represented as a Directed Acyclic Graph (DAG). Each node represents a financial feature, and edges represent probabilistic dependencies between them. Conditional Probability Tables (CPTs) are learned from historical data to quantify relationships between variables.

3.4 Details of Hardware & Software

The development and execution of the AlphaForge system require a combination of hardware and software components to ensure efficient data processing, model execution, and user interaction. The system is designed to be scalable and can operate on both local machines and cloud-based environments.

Hardware Requirements

The system does not require specialized hardware but performs efficiently on a standard computing setup. The minimum hardware requirements include a system with at least an Intel i5 processor or equivalent, 8 GB RAM, and sufficient storage capacity of around 10–20 GB for datasets and model files. A higher configuration, such as 16 GB RAM and SSD storage, is recommended for improved performance, especially during model training and backtesting.

The system is platform-independent and can run on Windows, Linux, or macOS environments. Internet connectivity is required for fetching real-time stock market data from external APIs.

Software Requirements

The AlphaForge platform is built using modern software technologies that support scalability, performance, and modularity.

The backend is developed using **FastAPI**, a high-performance Python framework used for building RESTful APIs. It handles data processing, feature engineering, and probabilistic inference. The core AI module is implemented using Python libraries such as **pgmpy** for Bayesian Networks and **networkx** for graph representation.

The frontend is developed using **Next.js**, a React-based framework that provides a responsive and interactive user interface. It supports dynamic dashboards, data visualization, and smooth user experience.

The system uses **Node.js** as the runtime environment for frontend development and package management. For data collection, the **yfinance API** is used to fetch historical and real-time stock market data.

For data storage, **Parquet files** are used as an offline feature store due to their efficient columnar storage format. Additionally, **Redis** is used as an in-memory data store for caching and fast retrieval of frequently accessed data.

The development environment includes tools such as Visual Studio Code for coding, Git for version control, and Python virtual environments for dependency management.

Experiment and Results

1 Dashboard Output

Description:

The dashboard provides a comprehensive overview of current market conditions by displaying key indicators such as market regime, volatility index, and active trading signals. It aggregates real-time data and presents it in a visually intuitive format for quick decision-making. The system categorizes the market into bullish, bearish, or sideways regimes based on computed features. This interface enhances user understanding of market trends and supports efficient analysis.

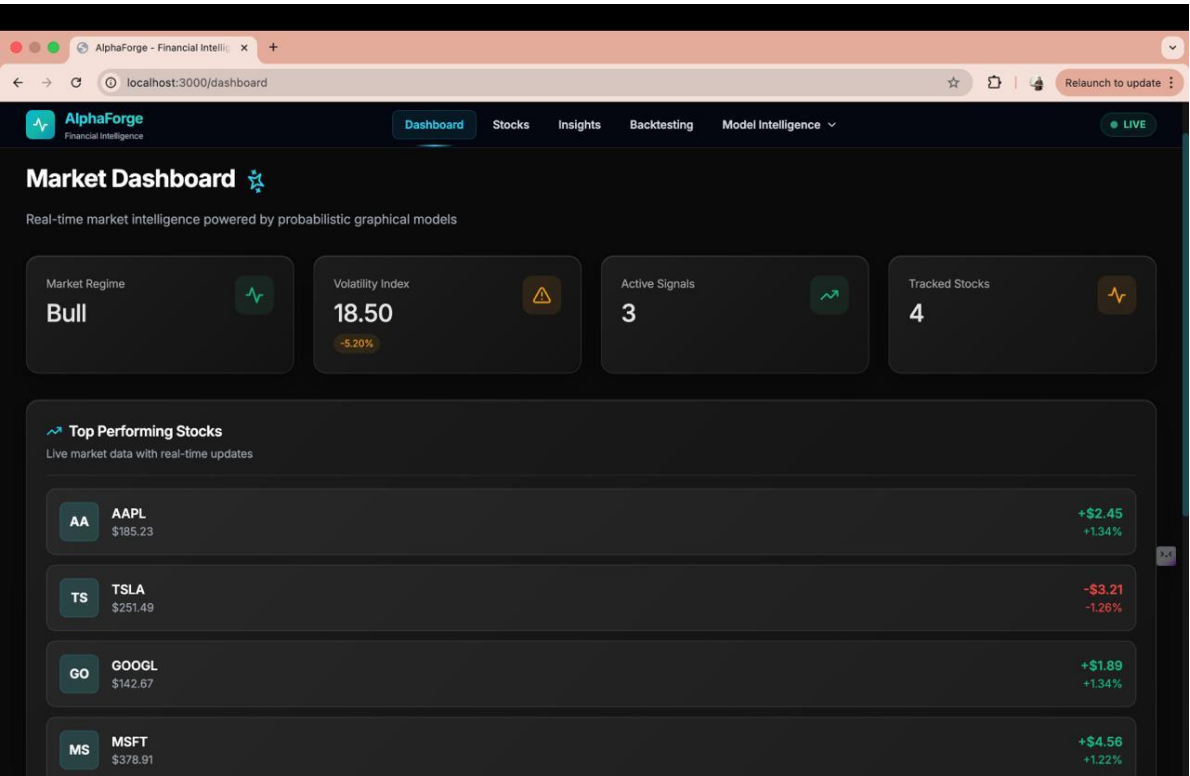


Figure 2: Market Dashboard displaying market regime, volatility index, and active signals

2. Stock Analysis Page

Description:

This figure illustrates detailed stock analysis for a selected company, including price trends and multiple technical indicators. The chart displays moving averages along with RSI and MACD indicators to identify momentum and trend strength. Volume data is also included to assess trading activity. These indicators collectively help in understanding price behavior and potential market movements.

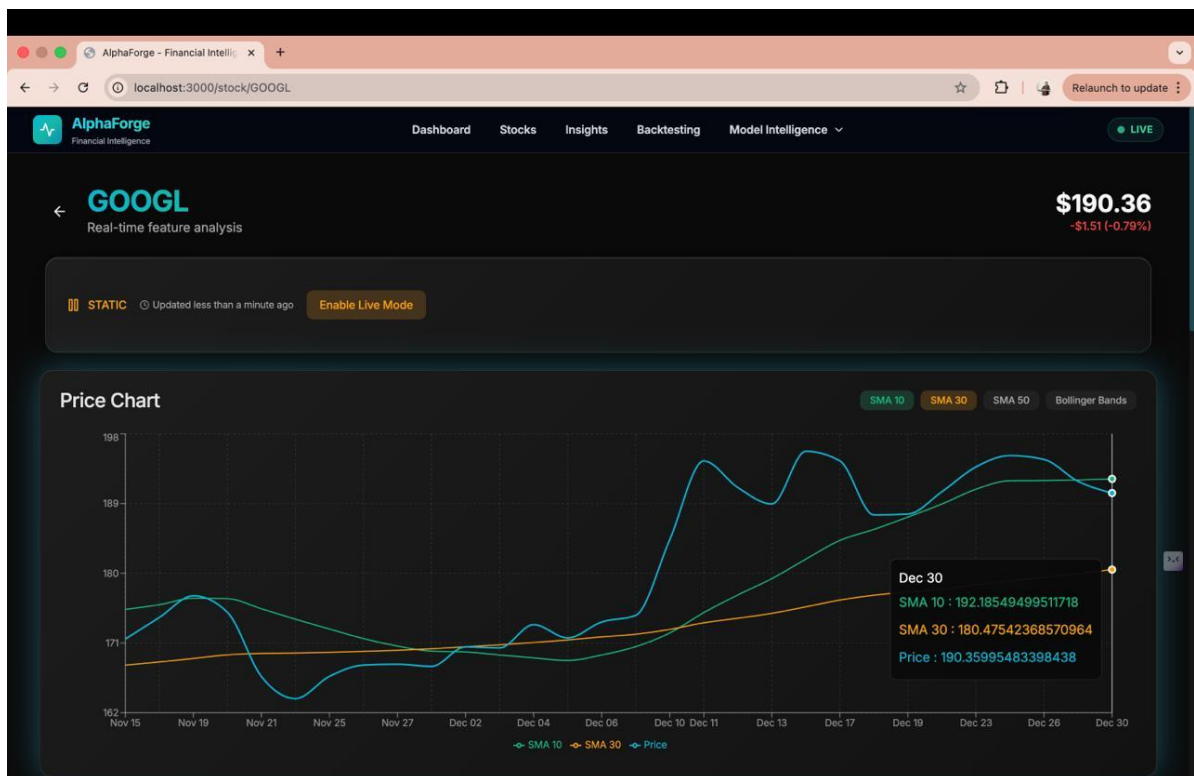


Figure 3: Stock analysis with price trends and technical indicators (SMA, RSI, MACD)

3. Backtesting Results

Description:

The backtesting module evaluates the performance of various trading strategies using historical data. Metrics such as Sharpe ratio, win rate, and total return are used to compare strategies. The results demonstrate the effectiveness of different approaches under varying market conditions. This helps in selecting optimal strategies for real-world trading scenarios.

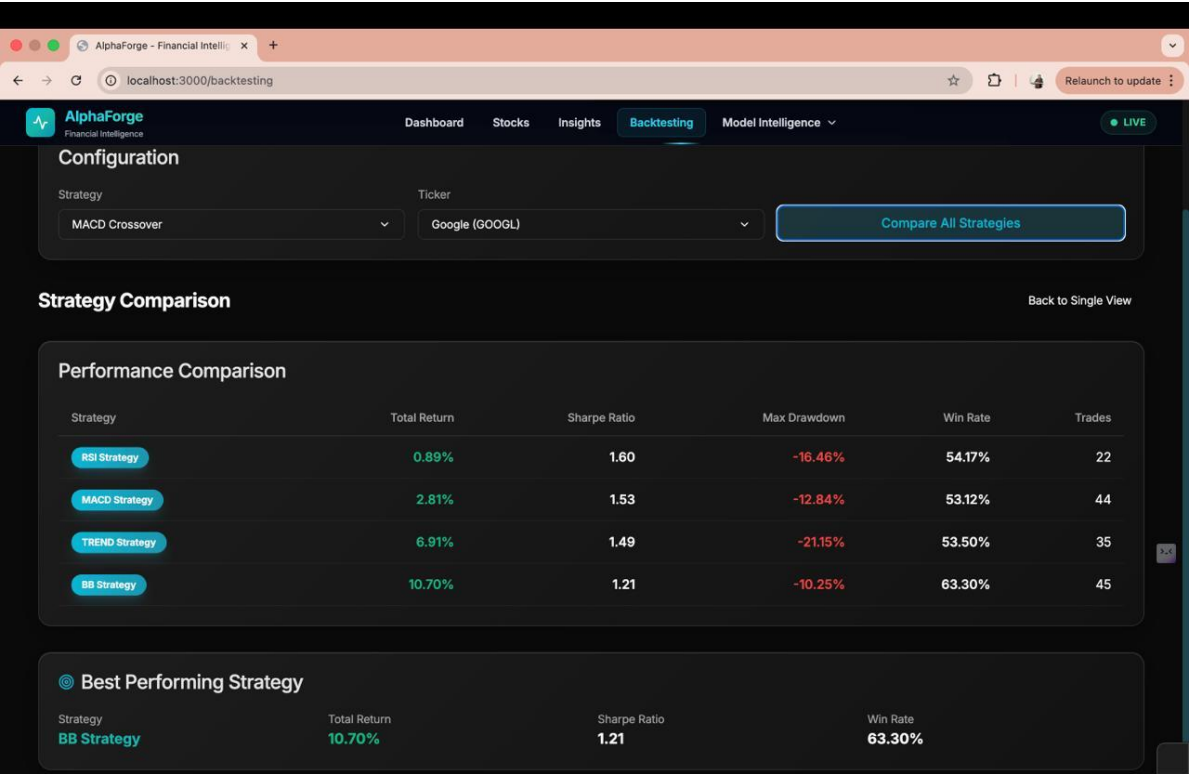


Figure 4: Backtesting performance comparison of trading strategies

4. Bayesian Network Visualization

Description:

This figure represents the Bayesian Network structure used in the system, modeled as a Directed Acyclic Graph (DAG). Each node corresponds to a financial feature, while edges indicate probabilistic dependencies between variables. The network captures causal relationships and enables probabilistic inference. It forms the core of the explainable AI model used for prediction.

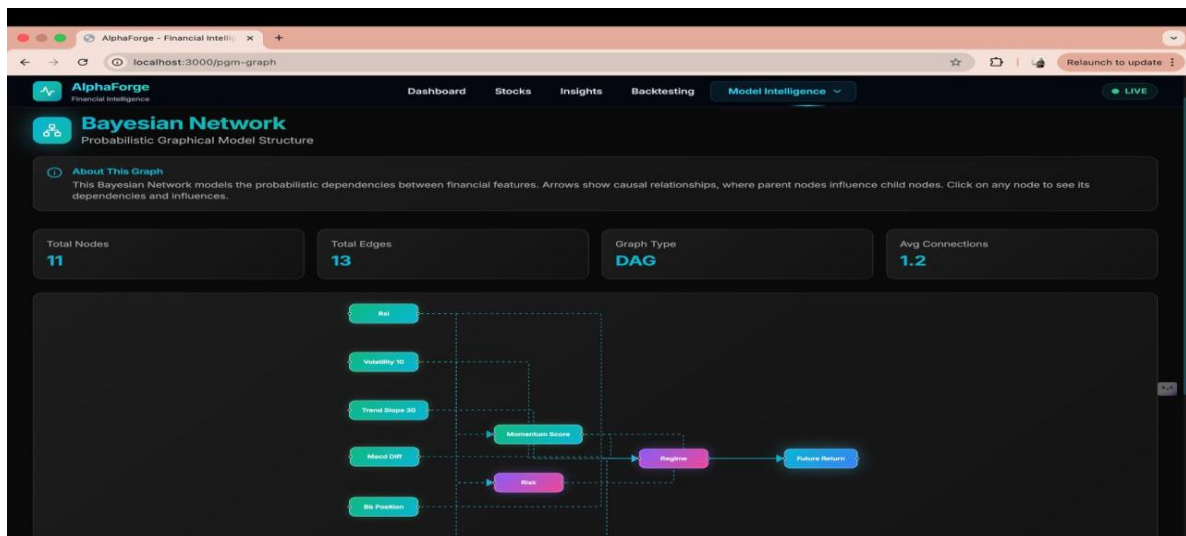


Figure 5: Directed Acyclic Graph (DAG) representing Bayesian Network

5. Structure Analysis

The structure analysis provides validation of the Bayesian Network by examining relationships between variables. It includes correlation analysis and dependency checks to ensure model accuracy. This step helps in identifying strong and weak connections between features. The analysis improves the reliability and interpretability of the model

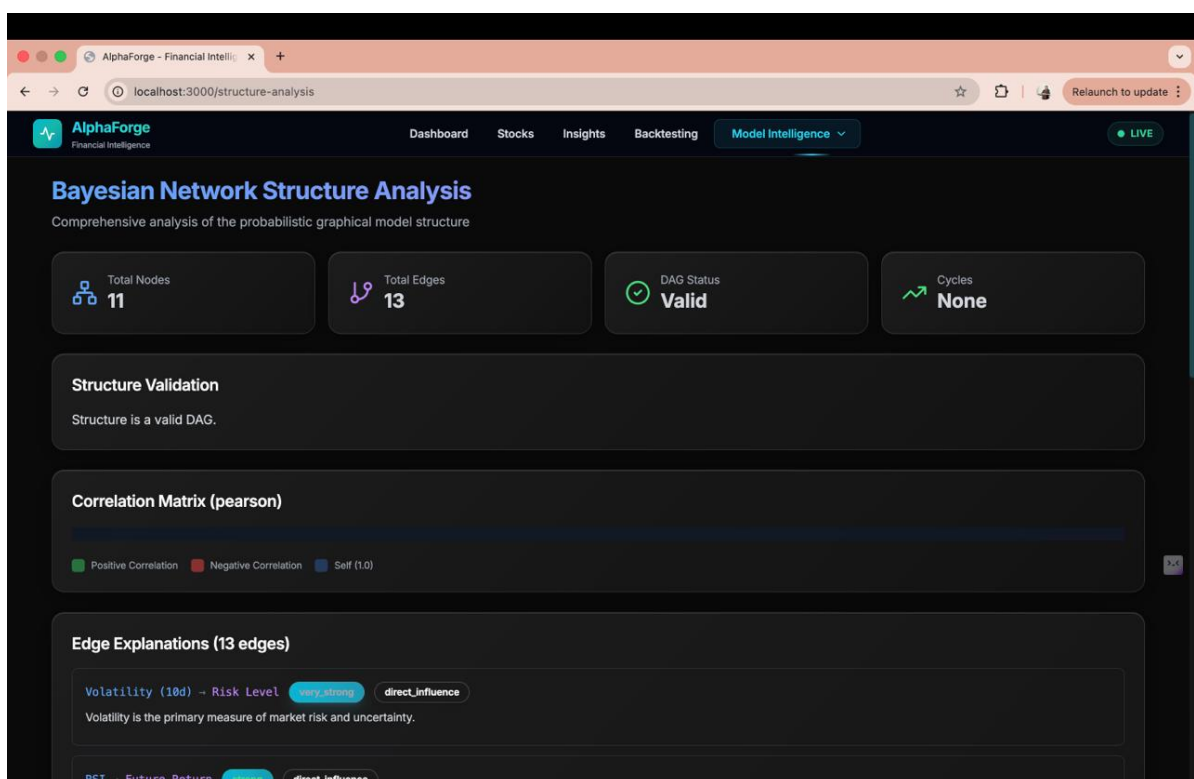


Figure 6: Bayesian Network structure validation and correlation analysis

6. Feature Contribution Analysis

This figure shows the contribution of individual features to the model's predictions. Each feature's impact is quantified and visualized using bar graphs. It highlights which indicators have the most influence on the predicted outcome. This improves transparency and helps users understand the reasoning behind predictions.

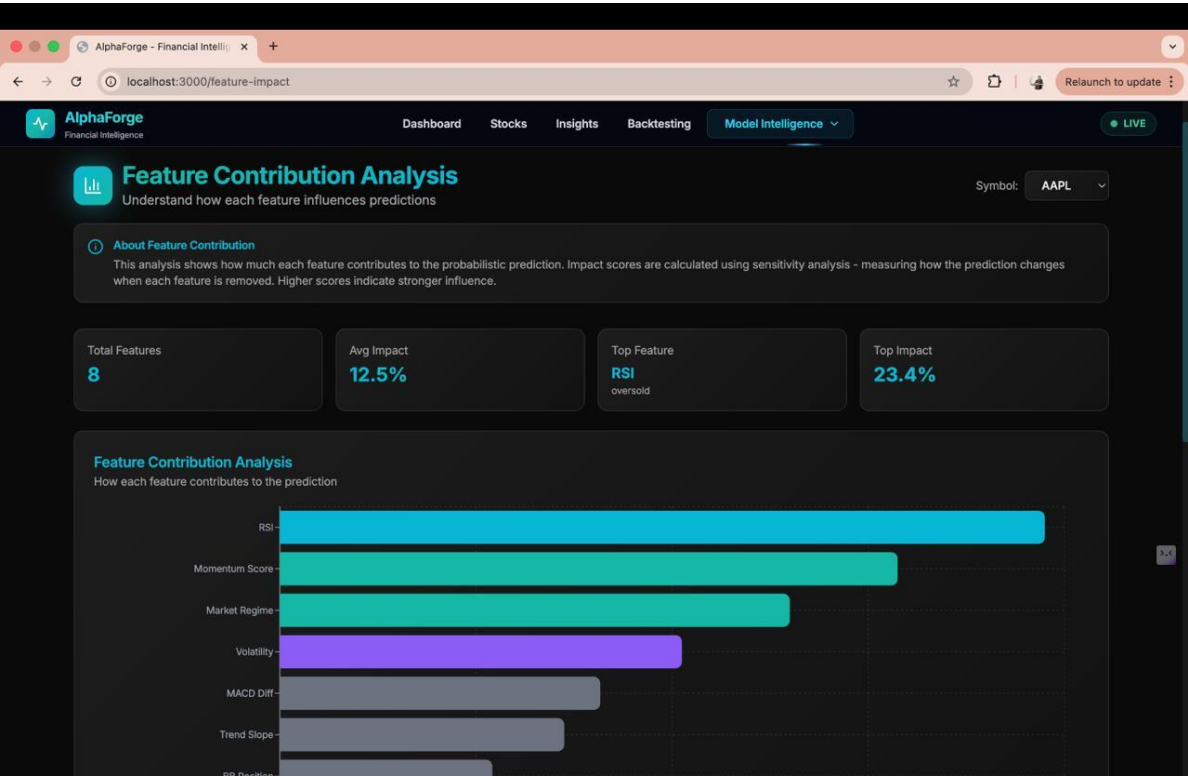


Figure 7: Feature contribution analysis showing impact of indicators on prediction

7. Model Comparison

The model comparison evaluates the performance of the PGM-based model against baseline models. Metrics such as accuracy and prediction quality are compared across different approaches. The results indicate that the proposed model achieves better interpretability while maintaining competitive performance. This validates the effectiveness of the system.

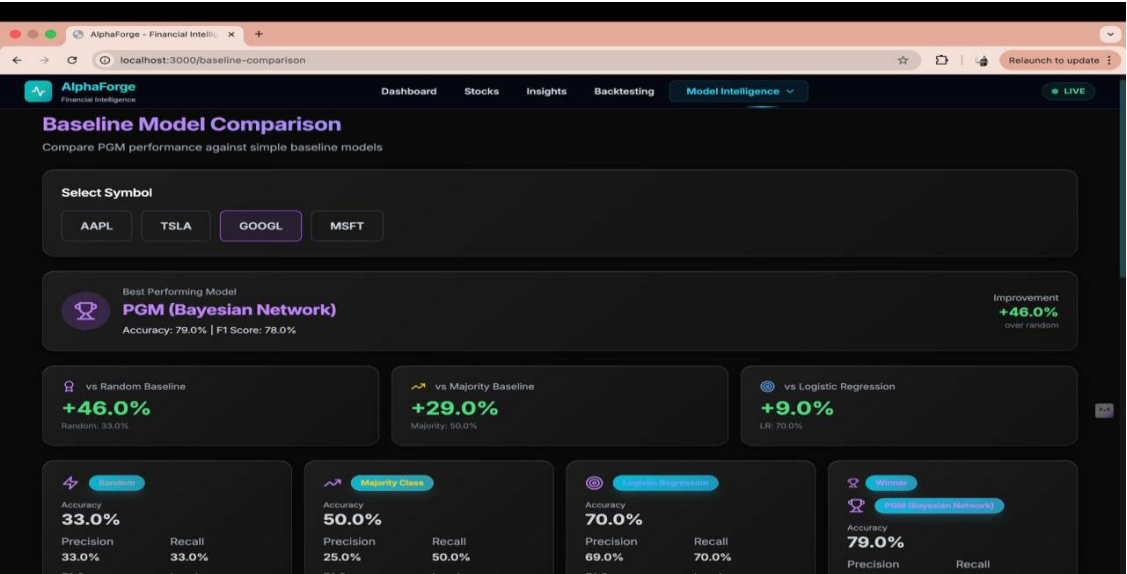


Figure X: Performance comparison of PGM model with baseline models

3.6 Conclusion and Future work

Conclusion

The AlphaForge system successfully demonstrates the application of Probabilistic Graphical Models in financial market prediction with a strong emphasis on explainability and transparency. Unlike traditional black-box models, the system provides probabilistic predictions along with clear interpretations of the contributing factors. The use of Bayesian Networks enables modeling of causal relationships between financial indicators, improving both prediction quality and user trust.

The integration of real-time feature engineering, interactive dashboards, and backtesting modules makes the system comprehensive and practical for real-world use. Experimental results indicate that the model achieves satisfactory accuracy while maintaining well-calibrated probability estimates. The feature contribution analysis and scenario simulation further enhance the system's ability to support informed decision-making.

Overall, the project successfully bridges the gap between predictive performance and interpretability, making it a valuable tool for traders, analysts, and researchers.

Future Work

Although the system performs effectively, several enhancements can be implemented to improve its capabilities further. One potential improvement is the integration of deep learning models such as LSTM or Transformer-based architectures to enhance predictive accuracy, especially for long-term dependencies.

The system can also be extended to support real-time streaming data using technologies such as Apache Kafka, enabling continuous model updates and live predictions. Deployment on cloud platforms such as AWS or Google Cloud can improve scalability and accessibility.

Additional features such as portfolio optimization, risk management modules, and sentiment analysis using news and social media data can further enrich the platform. Developing a mobile application using React Native can enhance user accessibility. Furthermore, incorporating automated model tuning and reinforcement learning-based strategies can significantly improve performance.

4 References

Sr. No.	Resource / Title	Link
1	pgmpy Documentation	https://pgmpy.org/
2	yfinance API Documentation	https://pypi.org/project/yfinance/
3	FastAPI Documentation	https://fastapi.tiangolo.com/
4	Next.js Official Documentation	https://nextjs.org/docs
5	Redis Documentation	https://redis.io/documentation

Sr. No.	Resource / Title	Link
6	Pandas Documentation	<u>https://pandas.pydata.org/docs/</u>
7	NumPy Documentation	<u>https://numpy.org/doc/</u>
8	Matplotlib Documentation	<u>https://matplotlib.org/stable/contents.html</u>
9	Investopedia (Technical Indicators Reference)	<u>https://www.investopedia.com/</u>
10	Bayesian Networks (Stanford Encyclopedia of Philosophy)	<u>https://plato.stanford.edu/entries/bayes- net/</u>
11	Scikit-learn Documentation	<u>https://scikit-learn.org/stable/</u>
12	Python Official Documentation	<u>https://docs.python.org/3/</u>